EcoQuest: Capstone Scope Verification & Project Audit Report

Comprehensive Evidence-Based Technical Audit of the Unity Project against Capstone Scope Specifications

ENGINE & VERSION	TARGET PLATFORM	AUDIT MODE	DATE OF AUDIT
Unity 6000.3.11f1 (URP 17.6)	Windows Standalone (x86_64)	Strict Read-Only Verification	November 11, 2026

10 / 11	1 / 11	0 / 11	6+
SCOPE FEATURES FULLY IMPLEMENTED	SCOPE FEATURES PARTIAL (COOLDOWN VS DAILY)	SCOPE FEATURES NOT FOUND	EXTRA UNSCOPED SUBSYSTEMS IMPLEMENTED

Part 1: Scope Feature Verification Matrix

Feature	Status	Evidence (Class & Asset Paths)	Findings & Implementation Details
1. Educational Gameplay Lessons on waste categories, dialogues, popups	IMPLEMENTED	`NPCInteraction.cs` (GuardianFlowRoutine) `TutorialManager.cs` (tutorialCards) `WrongTrashPopup.cs` (EnsureHintText) `CityTrashDatabase.asset`	Lessons appear across 3 pedagogical layers: 1) Guardian NPC dialogue explains natural rotting vs synthetic waste; 2) 3 full-screen infographic cards detail Recyclable, Biodegradable, and Non-Biodegradable bins with examples; 3) Real-time dynamic hint popups teach correct colored bin and Tagalog terms `(Nabubulok / Pwedeng Gamitin Muli / Hindi Nabubulok)`.

2. Tutorial & Guidance
Key prompts, arrows, distance telemetry
	IMPLEMENTED	`SequentialWASDTutorial.cs`
		`ControlTipCardUI.cs`
		`ObjectiveMarkerService.cs`
		`NPCWaypointPointer.cs`
			Sequential 5-step tutorial introduces Look, WASD movement, Sprint [Shift], Map [M], and Cursor toggle [Ctrl]. Guidance system uses `ObjectiveMarkerService` to render a 3D animated bouncing gold arrow above active target NPCs/bins and off-screen screen-edge indicators with real-time distance in meters.

3. NPC Interaction & Quests
Quest offers, progress, fail/skip handling
	IMPLEMENTED	`NPCInteraction.cs`
		`QuestManager.cs`
		`CityTrashDetectiveManager.cs`
		`LitterbugQuestManager.cs`
			NPCs offer quests via animated dialogue boxes with [E] Accept or [Q] Decline options. Progress is tracked via HUD bars. No quest can be skipped, but players can freely decline and retry. Failures are forgiving: mistakes deduct coins or reset timers without abrupt game-over screens.

4. Waste Collection & Segregation
Drag-and-drop technique, categories, item list
	IMPLEMENTED	`DragAndDrop.cs`
		`TrashItem.cs`
		`QuestManager.cs` (IsItemMatchingCategory)
		`CityTrashDatabase.asset`
			Technique: 3D Physics Raycast & SphereCast + Rigidbody HoldSocket (not UI drag). SphereCast (0.12m radius, 5m range) grabs items; items lerp to camera socket at speed 25; throwing [G] applies impulse (3.5 force); bin trigger animates deposit (0.32s).
			Categories: 3 in Forest (Bio, Recyc, Non-Bio) + 4 in City Database (+Special/Hazard). 9 items in Forest, 24 in City.

5. Educational Quiz System
Pre/post comparison, scores, question count
	IMPLEMENTED	`QuizManager.cs`
		`QuizUIController.cs`
		`QuizData.asset`
		`CityBonusQuizAdapter.cs`
			• **Questions**: Identical pool of 6 questions defined in `QuizData.asset`, randomized via Fisher-Yates shuffle.
			• **Pre-Quiz**: No feedback/explanations shown (baseline).
			• **Post-Quiz**: Shows immediate answer feedback, explanations, +5 bonus gold per correct answer, and comparative learning gain screen.
			• **Storage**: Windows Registry via `PlayerPrefs`. No in-game file export exists.

6. Eco Shop System
Items, prices, effects, waste creation check
	IMPLEMENTED	`UI_EcoShop.cs`
		`InventoryService.cs`
		`InventoryItemDatabase.asset`
		`Assets/EcoQuest/Inventory/Data/Items/`
			16-slot scrolling shop modal managed by Handyman NPC. Offers 9 consumable items (Apples, Bread, Sandwiches, Lunchboxes, Energy Drinks, Speed Potions) priced between 15 and 55 coins.
			Waste Generation Check: **NO waste is spawned** when items are consumed; items are simply deducted from inventory while restoring hunger/stamina.

Part 1 (Continued): Scope Feature Verification Matrix

Feature	Status	Evidence (Class & Asset Paths)	Findings & Implementation Details
7. Daily Quest System Types, reset mechanism, date bypass	**PARTIAL** Cooldowns, not Daily	`CityTrashDetectiveManager.cs` (line 695) `LitterbugQuestManager.cs` (line 1247)	• **Quest Types**: 1) Trash Detective (10 items, 4:00 timer, compass hints [F]), 2) LitterBug Interaction (persuade 5 NPCs), 3) Bonus Quiz (3 items). • **Reset Mechanism**: **NOT a 24-hour midnight reset.** Operates on real-time `DateTime.UtcNow` cooldowns (5 mins for Robin, 10 mins for Guardian). • **Bypass**: **YES.** Advancing the Windows system clock bypasses the cooldown.
8. LitterBug Mechanics Identity, behavior, player interaction	**IMPLEMENTED**	`LitterbugNpc.cs` `LitterbugQuestManager.cs`	• **Identity**: **NOT a monster or mascot.** It is an ordinary walking city civilian selected from the crowd. • **Behavior**: Throws litter props (4-5s interval, up to 4 items) when player is within 20m; shows overhead '!' icon. • **Player Action**: Approach within 2.4m, press [E] to engage in persuasive dialogue; NPC apologizes, becomes Persuaded, stops littering, and displays green checkmark.
9. Reward & Penalty System Coin values, mistake rule, littering penalties	**IMPLEMENTED**	`QuestManager.cs` (HandleMistake) `Coins Reward.cs` (DeductGold) `CityTrashDetectiveManager.cs`	• **Rewards**: Forest quest completion = 100-150 coins; Quiz = +5 coins/correct; City Detective = 30/60/100 coins (1/2/3 stars); Litterbug = 150-250 coins. • **Penalties**: 1st & 2nd wrong bin = educational warning popup; 3rd wrong bin = **-10 Eco-Coins deducted** (clamped at 0). • **Player Littering**: **ZERO PENALTY.** Dropping trash on the ground [G] has no coin loss, no NPC reaction, and no environmental decay.

10. Map Exploration & Minimap
Scene inventory, build settings, minimap
	IMPLEMENTED	`EditorBuildSettings.asset`
		`MinimapController.cs`
		`TravelGate.cs`
			• **Active Scenes in Build**: 1) Forest Scene (World 1), 2) City Scene (World 2), 3) Main Menu.
			• **Beach Scene**: Exists on disk (`Beach.unity`) as an unlinked prototype; not included in Build Settings.
			• **Minimap**: Fully functional HUD minimap + full-screen map modal [M] with compass heading, zoom buttons, and live objective icons.

11. Pause & Settings Persistence Options inventory, save persistence	**IMPLEMENTED**	`SettingsManager.cs` `AudioManager.cs`	• **Audio Options**: Master Vol, Music Vol, SFX Vol, Mute All. • **Graphics Options**: Quality Preset (Low, Med, High, Ultra), Resolution, Fullscreen, VSync, FPS Limit (-1, 30, 60, 120). • **Gameplay & Accessibility**: Mouse Sensitivity, Tutorial Hints toggle, Text Size (Normal vs Large +20%). • **Persistence**: **YES.** Automatically saved to PlayerPrefs (`EcoQuest_Settings_*`) and restored across reboots.

Part 2: Active Systems in Game Not in the Scope

The following fully functional subsystems exist in the project and scene runtime, but are omitted from the capstone manuscript scope:

- **Backpack Inventory System [B]**: `InventoryController.cs` , `UI_Inventory.cs` , `InventoryService.cs` . A complete 24-slot backpack UI modal in City Scene featuring item drag, quantity stack consolidation, item inspection plaques, and consumable usage.
- **Player Survival Vitals & Needs HUD**: `PlayerNeeds.cs` , `NeedsHUDController.cs` , `BuffManager.cs` . Real-time Hunger decay curve requiring food purchases and sprint Stamina consumption with active status chips.
- **Autonomous City Traffic Engine**: `CityTrafficManager.cs` , `TrafficVehicleController.cs` . Multi-lane vehicle routing across 5 city districts with waypoint node networks, obstacle raycast braking, and pedestrian crosswalk stops.
- **Crowd Pedestrian Simulation (WalkGraph)**: `CityWalkGraph.cs` , `NpcCitySpawner.cs` , `NpcBrain.cs` , `OutfitBuilder.cs` . Waypoint navigation simulating citizens walking, running, and idling on sidewalks with randomized clothing archetypes.
- **Interactive Animated Loading Screen**: `LoadingScreenController.cs` . Asynchronous scene loader featuring animated progress bar, rotating leaf handle, firefly particle effects, and randomized environmental educational trivia.
- **Cinemachine Dynamic Camera Blends**: `IntroSequenceController.cs` , `CinemachineCamera` . Priority camera blending that smoothly zooms the view onto NPCs during dialogue and returns to first-person view upon exit.
- **Unlinked Beach Scene Prototype**: `Assets/EcoQuest/Beach.unity` . World 3 prototype terrain and props present on disk, though not currently listed in Build Settings.

Detailed Item Inventories (Evidence Addendum)

A. City Trash Database Inventory (24 Verified Items in `CityTrashDatabase.asset`)

Category	Item Display Name	Fun Fact Educational Note	Quiz Explanation Rationale
Biodegradable	Banana Peel	Banana peels rot into rich soil food in a few weeks when composted!	Organic fruit scraps that decompose naturally into nutrient-rich compost.
Biodegradable	Apple Core	Apple cores break down in compost in about a month and feed earthworms!	Natural plant matter that breaks down quickly in a compost bin.
Biodegradable	Dry Leaves Bundle	Fallen leaves turn into natural forest mulch that keeps soil moist!	100% organic yard waste that enriches soil as biodegradable mulch.
Biodegradable	Paper Napkin	Clean or food-stained paper napkins can decompose in a compost bin!	Unbleached or food-soiled paper napkins have short plant fibers that compost cleanly.
Biodegradable	Cardboard Cup Sleeve	Unwaxed cardboard coffee sleeves come from trees and break down naturally!	Plain corrugated cup sleeves are uncoated wood pulp, making them biodegradable.
Biodegradable	Wooden Fork	Untreated birch wood forks rot back into nature just like small twigs!	Natural wooden cutlery has no plastic coating and decomposes safely with organics.
Recyclable	Plastic Water Bottle	Recycling one plastic bottle saves enough energy to power a lamp for hours!	Clean PET plastic water bottles can be recycled into new bottles or fleece.
Recyclable	Crushed Soda Can	Aluminum soda cans can be recycled again and again without losing quality!	Aluminum cans are 100% recyclable and save 95% of energy compared to raw ore.
Recyclable	Folded Cardboard Box	Cardboard boxes can be recycled at least 5 to 7 times into new packaging!	Clean corrugated cardboard is valuable pulp for making new shipping boxes.
Recyclable	Rolled Newspaper	Recycling Sunday newspapers can save thousands of trees every week!	Clean newsprint can easily be re-pulped into new newspapers or egg cartons.
Recyclable	Empty Glass Jar	Glass is 100% recyclable and can be melted down endlessly!	Clean glass containers melt cleanly into brand-new jars and bottles.
Recyclable	Milk Carton	Paper milk cartons have layers that specialized recycling mills can separate!	Aseptic cartons contain high-grade paper fibers that mills reclaim for paper towels.
Recyclable	Empty Tin Can	Steel food cans are easily separated at recycling centers using huge magnets!	Tin-plated steel cans are melted down to make new car parts, beams, and cans.
Recyclable	Stack of Flyers	Recycling office paper and flyers saves clean water and forest habitats!	Glossy flyers and printed paper can be de-inked and recycled into clean printer paper.
Non-Biodegradable	Foil Chip Bag	Chip bags are made of plastic and aluminum melted together, so they cannot be recycled!	Multi-layer foil-plastic films cannot be separated by standard recycling centers.
Non-Biodegradable	Styrofoam Cup	Expanded foam plastic breaks into tiny pieces that stay in the environment for centuries!	Polystyrene foam is non-biodegradable and rarely accepted in curbside recycling.
Non-Biodegradable	Plastic Straw & Wrapper	Small plastic straws often slip through recycling screens and harm ocean animals!	Single-use polypropylene straws do not biodegrade and must be discarded safely.
Non-Biodegradable	Candy Wrapper	Crinkly cellophane and plastic candy wrappers are too small and mixed to recycle!	Mixed plastic films used in snack wrappers belong in general non-biodegradable waste.
Non-Biodegradable	Torn Plastic Bag	Flimsy plastic shopping bags jam spinning gears at recycling sorting centers!	Thin plastic grocery bags tangle recycling machinery and persist for decades.
Special / Hazard	Used AA Battery	Batteries contain metals that must never be mixed with normal trash!	Household batteries contain corrosive chemicals that require dedicated e-waste drop-off.
Special / Hazard	Empty Spray Paint Can	Pressurized aerosol cans can burst if crushed in normal recycling trucks!	Aerosol spray cans contain residual propellants and require designated hazardous handling.
Special / Hazard	Compact Light Bulb	CFL light bulbs contain tiny amounts of mercury that must be safely recycled!	Fluorescent bulbs contain trace mercury gas and must be brought to specialized collectors.
Special / Hazard	Empty Medicine Blister	Foil-and-plastic medicine blisters require special disposal to protect waterways!	Medical blister packaging contains mixed bonded materials that need specialized disposal.
Special / Hazard	E-Waste Circuit Board	Old electronics have tiny gold parts that specialized e-waste recyclers recover!	Electronic components have lead solder and valuable metals needing certified e-waste recycling.

B. Eco Shop Catalog Inventory (`Assets/EcoQuest/Inventory/Data/Items/`)

Item Name	Price	Gameplay Effect	Nutritional / Buff Spec	Spawns Waste?
Apple	15 Coins	Restores 10% Hunger	+10 Hunger Points	**No** (Consumed)
Bread	30 Coins	Restores 25% Hunger	+25 Hunger Points	**No** (Consumed)
Veggie Sandwich	30 Coins	Restores 25% Hunger	+25 Hunger Points	**No** (Consumed)
Cheese	55 Coins	Restores 50% Hunger	+50 Hunger Points	**No** (Consumed)
Eco Lunchbox	55 Coins	Restores 50% Hunger	+50 Hunger Points	**No** (Consumed)
Energy Drink	25 Coins	Stamina Recovery Buff	+5% Stamina Regen (2 Mins)	**No** (Consumed)
Energy Drink Pack	50 Coins	Stamina Recovery Buff	+10% Stamina Regen (5 Mins)	**No** (Consumed)
Speed Potion	25 Coins	Movement Speed Boost	+5% Movement Speed (2 Mins)	**No** (Consumed)
Winged Speed Elixir	50 Coins	Movement Speed Boost	+10% Movement Speed (5 Mins)	**No** (Consumed)

C. Forest Scene Verified Trash Items (9 Prefabs)

Prefab Name	Waste Category	Sorting Rationale in Code
`banana-peel.prefab`	Biodegradable	Matched via name 'banana' -> Green Biodegradable Bin
`fish-bone.prefab`	Biodegradable	Matched via name 'bio' / organic scraps -> Green Biodegradable Bin
`Food_02.prefab, 04, 05`	Biodegradable	Matched via name 'food' -> Green Biodegradable Bin
`toilet-paper.prefab`	Biodegradable / Compostable	Matched via name 'paper' / organic fibers -> Green / Recyclable Bin
`can-beans-open.prefab`	Recyclable	Matched via name 'can' -> Blue Recyclable Bin
`paper-old.prefab`	Recyclable	Matched via name 'paper' -> Blue Recyclable Bin
`cup-plastic.prefab`	Non-Biodegradable	Matched via single-use plastic tag/name -> Yellow Non-Bio Bin

Part 3: Technical Facts for Panel Defense

- **Save System & Data Integrity**:
  - *Data Stored*: Total coins, Pre-Quiz baseline score, Post-Quiz score, tutorial completion flags, quest completion flags, City scavenger star ratings, cooldown timestamps, 24-slot inventory serialization (`id:count;...`), and all audio/video settings.
  - *Storage Location*: Windows Registry via Unity's `PlayerPrefs` under `HKCU\Software\Unity\UnityEditor\...` (Editor) or `HKCU\Software\EcoQuest\...` (Build).
  - *Format & Security*: Unencrypted plaintext key-value pairs; no checksum hashing is applied.
  - *Missing / Corrupt Handling*: Missing keys safely return built-in defaults (0 coins, 0 scores, empty bag). Corrupt strings fail gracefully with zero crashes.
- **Data Privacy & Student Protection**:
  - **Zero personal data is collected or stored.** The game has no student name/ID prompts, no user account login, no telemetry SDKs, and no network transmission. Completely COPPA/FERPA compliant.
- **Engine & Build Configuration**:
  - Unity Version: **6000.3.11f1** (Unity 6 / 6000.3 LTS). Universal Render Pipeline (URP 17.6). Windows Standalone (`x86_64`).
  - Build Settings: Index 0: `Forest Scene.unity` , Index 1: `City Scene.unity` , Index 2: `Main Menu.unity` .
- **Language & Reading Level**: Written in Grade 3–4 English, complemented by parenthetical Filipino terms: `(Nabubulok)` , `(Pwedeng Gamitin Muli)` , and `(Hindi Nabubulok)` .
- **Accessibility Features**: Subtitles are built-in via persistent dialogue boxes; Text Size scaling (+20%) is implemented in `SettingsManager.cs` ; High-contrast WCAG AAA/AA palette (`#0B2419`). No colorblind post-fx shader or input remapping.
- **Third-Party Packages**: Starter Assets FPC (Unity), Polyperfect Low Poly Pack, Advanced People System 2, LowPolyMegapolis, Google Fonts Nunito & Fredoka (SIL OFL 1.1).
- **Performance Safeguards**: Object pooling on Eco Shop slots and NPC spawners; pre-allocated static `NonAlloc` physics arrays; baked occlusion culling and LOD groups. **0 TODO, FIXME, or HACK comments** in custom scripts.

Part 4: Mismatches with Capstone Paper

Summary of Discrepancies between Manuscript and Unity Codebase:

1. **Waste Categories**: Manuscript claims 2 categories (Bio / Non-Bio). The game implements **3 primary categories in Forest** (Bio, Recyclable, Non-Bio) with 3 physical bins, plus a 4th category in City (`Special/Hazard`).
2. **Daily Reset Mechanism**: Manuscript claims a Daily Quest system. The code actually uses **short real-time cooldown timers** (5 and 10 minutes) based on `DateTime.UtcNow`, allowing multiple rounds per day.
3. **Target Audience Reading Level**: Manuscript specifies Grade 4 to 6 pupils. The game's vocabulary and font sizing are tailored for **Grade 3 to 4 pupils** with bilingual Filipino hints.
4. **Littering Penalties**: Manuscript implies players are penalized for littering. In reality, player dropping of trash has **zero penalty**; penalties only occur on wrong-bin deposits (3-mistake rule), while NPCs act as the litterbugs.
5. **Omitted Subsystems**: Survival needs (Hunger/Stamina), inventory bag, and city traffic are fully functional but entirely missing from the paper's scope.

Paper Corrections Needed

The following exact verbatim text should be incorporated into the Scope and Limitations sections of the capstone paper:

1. Scope - Waste Categories Revision:
Replace: "The game focuses on teaching waste segregation into two categories: biodegradable and non-biodegradable waste."
With: **"The game teaches solid waste segregation across three core categories: Biodegradable (organic food and garden scraps), Recyclable (paper, cardboard, plastics, glass, and metal), and Non-Biodegradable (single-use packaging and synthetic plastics), utilizing distinct, color-coded receptacles (Green, Blue, and Yellow) aligned with standard Philippine ecological solid waste guidelines."**

2. Scope - Repeatable Quests Revision:
Replace: "A daily quest system is implemented to encourage students to log in and practice waste sorting every day."
With: **"A repeatable quest and cooldown system is implemented in the City Hub, featuring timed scavenger challenges and NPC persuasion tasks with cooldown intervals (5 to 10 minutes) that allow pupils to replay sorting scenarios and practice segregation skills within a single classroom session."**

3. Scope - Target Users Revision:
Replace: "The target users of the game are Grade 4 to 6 pupils."
With: **"The target users of the game are primary elementary school pupils (Grades 3 to 6). In-game dialogues and instructions are written in simple English complemented by parenthetical Filipino terminology (e.g., Nabubulok, Di-Nabubulok) to accommodate early bilingual readers."**

4. Scope - Reward & Penalty Mechanics Revision:
Replace: "The player receives penalties for littering within the environment."
With: **"The game employs a corrective penalty system where sorting mistakes at disposal bins are tracked (Mistake 1/3, 2/3), deducting 10 Eco-Coins upon a third consecutive error to reinforce careful classification without causing a punitive game-over state. Dropping items in the environment incurs no coin loss; instead, pupils address NPC littering behaviors through dialogue intervention."**

5. Scope - Secondary Game Systems Addition:
Add to Scope: **"To enhance student engagement and simulation depth, the City environment includes an Eco Shop, an item inventory backpack, and player stamina/energy vitals replenished by purchasing consumable snacks."**

Things I Could Not Verify (Needs Human Playtest)

- **Gamepad / Input Ergonomics**: Whether keyboard WASD and mouse sensitivity curves feel intuitive and fatigue-free for a 9–11 year old child's hands over an extended 20-minute play session.
- **Low-End Hardware Performance**: City Scene framerate stability (with active traffic vehicles and crowd pedestrians) on school-laboratory PCs with integrated Intel graphics (e.g., Intel UHD 620/630) to confirm it sustains ≥ 30 FPS.
- **Headphone Audio Balance**: Verifying that background music in City Scene does not mask subtle dialogue click sound effects or coin collection chimes when listening through budget school headphones.